

ACADEMIC YEAR: 2026-27(ODD)

SEMESTER: 3

SUBJECT: 102040306 – INTRODUCTION TO PYTHON PROGRAMMING

PRACTICAL LIST

Sr. No.	Name of Experiment
1	A. Write a Python program to accept two numbers from the user and perform addition, subtraction, multiplication, and division. B. Write a Python program to calculate the area and circumference of a circle by accepting the radius from the user. C. Write a Python program to convert temperature from Celsius to Fahrenheit and Fahrenheit to Celsius based on user choice.
2	A. Write a Python program to check whether a given number is positive, negative, or zero. B. Write a Python program to find the largest of three numbers entered by the user using nested if-else statements. C. Write a Python program to calculate the grade of a student based on marks entered by the user.
3	A. Write a Python program to generate the Fibonacci series up to n terms entered by the user. B. Write a Python program to find the factorial of a given number using a while loop. C. Write a Python program to display all prime numbers between two given numbers.
4	A. Write a Python program to define a function that checks whether a number is a palindrome. B. Write a Python program to define a function that calculates simple interest using parameters and return values. C. Write a Python program to define a function that returns the maximum and minimum values from a list. D. Write a Python program to compute the factorial of a given number using a recursive function.
5	A. Write a Python program to create a list of n integers and perform insertion and deletion operations on the list. B. Write a Python program to store elements in a tuple and access the tuple elements using indexing and looping statements. C. Write a Python program to create a dictionary with key-value pairs and perform basic operations such as insertion, deletion, and retrieval of values.
6	A. Write a Python program to perform string operations such as concatenation, length calculation, case conversion, and substring extraction. B. Write a Python program to read the contents of a text file using appropriate file handling modes and display them on the screen. C. Write a Python program to write data into a text file and append additional data to the same file using file handling operations.
7	A. Write a Python program to handle division by zero using try and except blocks. B. Write a Python program to handle multiple exceptions in a single program. C. Write a Python program to handle runtime errors using try, except, else, and finally blocks.
8	A. Write a Python program to create a class and an object and access class attributes and methods. B. Write a Python program to initialize class data using a constructor.

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	C. Write a Python program to implement a simple class-based application using object-oriented concepts.
9	A. Write a Python program to generate five random integers between 1 and 100 using the random module and calculate their square roots using the math module. B. Write a Python program to define a lambda function that accepts two numbers and returns the greater number. C. Write a Python program to create a list of integers and use built-in functions len(), max(), min(), and sum() to display the total number of elements, maximum value, minimum value, and sum of elements in the list.
10	A. Write a Python program to implement a stack using a list with operations such as push, pop, and display. B. Write a Python program to develop a small game where the computer generates a random number, and the user guesses the number within a limited number of attempts. C. Write a Python program to create a simple contact management system that allows adding new contacts, saving contact details to a text file, and retrieving and displaying the saved contacts.
11	Design and implement a mini-project using Python concepts covered in the course.